

# Refugee inclusion and the unmasking of AI models for the assistance of UNHCR operators

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## 1 Introduction and problem formulation

The purpose of this project is the implementation of an AI informed pipeline and dashboard for assisting UNHCR operators on the ground in the identification and selection refugee households eligible for economic assistance. The use of Machine Learning and Artificial Intelligence tools in humanitarian context is a growing practice [1], driven not only by increasing technological capabilities but also by growing needs for humanitarian assistance amid ongoing refugee crises and decreasing economic support from states. In a political context characterized by direct attacks and consequent withdrawal of funds devoted to humanitarian efforts<sup>1</sup>, humanitarian agencies have resorted to AI for decreasing operation costs and avoiding the overload and burnout of operators on the-ground [1]. However, while the benefits of AI have been outlined, ethical concerns over algorithmic bias, overreliance on recommendation systems, ‘techno-solutionist’ approaches [2], as well as risks of identification and targeting of vulnerable populations [5], have been raised. The need for human oversight over automated decision-making becomes not only crucial but also a legal requirement in situations where imperfect and potentially biased AI decisions have concrete impacts on human lives [4, 6, 3]. Difficulties in ensuring that the ‘humans-in-the-loop’ take responsibility in ‘overriding’ AI decisions are well known. Humanitarian operators - facing the double burden of being accountable in overriding AI systems (of which they have little technical knowledge) while independently assessing each case - risk simply accepting AI recommendations on the eligibility of vulnerable populations. For this reason, it is necessary to move from individually-centered to institutionally-centered frameworks of accountability [3]. Following this approach, we propose the implementation of an ‘algorithm-in-the-loop’ approach to humanitarian assistance [3], leveraging the potential of AI to learn from historical records of eligible/non-eligible households and guiding, rather than suggesting, decisions to UNHCR operators.

## 2 Leveraging the experiences of UNHCR operators to inform future choices

After an initial screening conducted through on-field interviews, humanitarian operators compile Scorecards where they describe refugees’ households through a series of indicators such as demographics, basic needs, housing status, dependency burden, language literacy. To this, they add a short recording summarizing the case, giving their first impressions and initial assessments right after the interview. The data are then used to assess the households’ vulnerability and their eligibility to economic assistance programmes. Records of previous decisions can be used to learn from the accumulated experience of UNHCR workers, assess households’ vulnerability and guide decision making AI. However, previous attempts at implementing AI systems have shown good results in assessing vulnerability but poor predictive power when it comes to eligibility. The choice of eligibility is in fact conditioned by other crucial factors such as what is said in the

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<sup>1</sup><https://www.theguardian.com/world/2025/jan/28/un-aid-cuts-trump-funding>.

summary recordings, the availability of budget, and expectations of future incoming refugees. In short, operators knowledge and contextual assessment is not fully captured by AI models.

In this context, ensuring accountability and human override is necessary. For this reason, we implement a strategy that aids UNHCR operators while removing the binding binary choice (eligible/non-eligible) suggested by AI. For this objective, we construct an AutoML model AutoGluon model similar to Cashy (**K-Mazzu**) and we repurpose it from prediction to feature extraction. Rather than returning a binary, we focus on the key predictors (the features) in the models (based on what has oriented the eligibility/non-eligibility choice in the past). We use these key indicators to assess, in each household profile, whether they manifest particular factors that have oriented previous choices.

These factors are then visualized in the operator’s dashboard, ordered by their predictive importance (but with no additional information on any predictive value of the indicators). In addition, the dashboard also allows operators to listen to the interview recording and still consult the full profile Scorecard, if needed. Only for those cases where the model would predict a very high risk case - a ‘flag’ appears signaling households at high vulnerability risk. The responsibility of the decision thus lies entirely in the operator - with no need to override automated decisions, but all key information needed for making an informed choice is provided. The key advantage of AI models - reducing costs and time - is thus obtained by reducing the complexity of scorecards to a minimum through K-Mazzu, while removing any bias introduced by AI-recommended choices and thus avoiding the need of monitoring AI override.

The proposed pipeline achieves the purpose of assisting UNHCR operators in making technology-informed but entirely human decisions over the assistance of vulnerable refugee households. It does so by reducing the complexity of each case assessment and thus fastening decision-making and lowering the work overload of humanitarian workers. In this way, the pipeline ensures the institutional accountability ‘by-design’, by implementing an ‘algorithm-in-the-loop’ approach [3]. Monitoring of improved performance can easily be assessed through each case processing time and workload and well-being surveys to humanitarian workers. A further advantage of the process is that it is highly context-aware: not only can the K-Mazzu algorithm be trained on location-specific historical records (thus allowing for the identification of context-specific factor salience), but the choices of operators will keep being informed by context-dependent and situational factors.

## References

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